

**DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE**

**Supplementary Summer Examination-2024**

**Course: B. Tech.                      Branch: Computer Engineering /CSE                      Semester: VII**

**Subject Code & Name: BTCOC701 Artificial Intelligence**

**Max Marks: 60                              Date: 01/07/2024                              Duration: 3 Hr.**

**Instructions to the Students:**

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in ( ) in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

|                                                                                 | (Level/CO) | Marks     |
|---------------------------------------------------------------------------------|------------|-----------|
| <b>Q. 1 Solve Any Two of the following.</b>                                     |            | <b>12</b> |
| A) What is Artificial Intelligence? Explain its future scope.                   | Remember   | 6         |
| B) Write down brief history and evolution of Artificial Intelligence.           | Understand | 6         |
| C) Explain in detail, the structure of different intelligent agents.            | Understand | 6         |
| <b>Q.2 Solve Any Two of the following.</b>                                      |            | <b>12</b> |
| A) Give the outline of uniform cost search algorithm with suitable example.     | Remember   | 6         |
| B) Write A* algorithm and discuss the various observations about algorithm.     | Remember   | 6         |
| C) Explain how the steepest accent hill climbing works and Heuristic functions? | Understand | 6         |
| <b>Q. 3 Solve Any Two of the following.</b>                                     |            | <b>12</b> |
| A) List and Explain the issues in knowledge representation.                     | Remember   | 6         |
| B) State representation of facts in propositional logic with an example.        | Understand | 6         |
| C) Differentiate between forward and backward reasoning.                        | Understand | 6         |
| <b>Q.4 Solve Any Two of the following.</b>                                      |            | <b>12</b> |
| A) Discuss briefly about Bayesian Network.                                      | Remember   | 6         |
| B) Explain about Dempster-shafer theory.                                        | Understand | 6         |
| C) List out the planning terminologies and components of planning.              | Remember   | 6         |

|             |                                                                                                         |                   |           |
|-------------|---------------------------------------------------------------------------------------------------------|-------------------|-----------|
| <b>Q. 5</b> | <b>Solve Any Two of the following.</b>                                                                  |                   | <b>12</b> |
| <b>A)</b>   | <b>What is Natural Language Processing? Explain various types of NLP techniques.</b>                    | <b>Remember</b>   | <b>6</b>  |
| <b>B)</b>   | <b>Comparison between Syntactic Processing &amp; Semantic Processing.</b>                               | <b>Understand</b> | <b>6</b>  |
| <b>C)</b>   | <b>With neat sketch, explain the architecture, characteristics features and roles of expert system.</b> | <b>Remember</b>   | <b>6</b>  |

**\*\*\* End \*\*\***